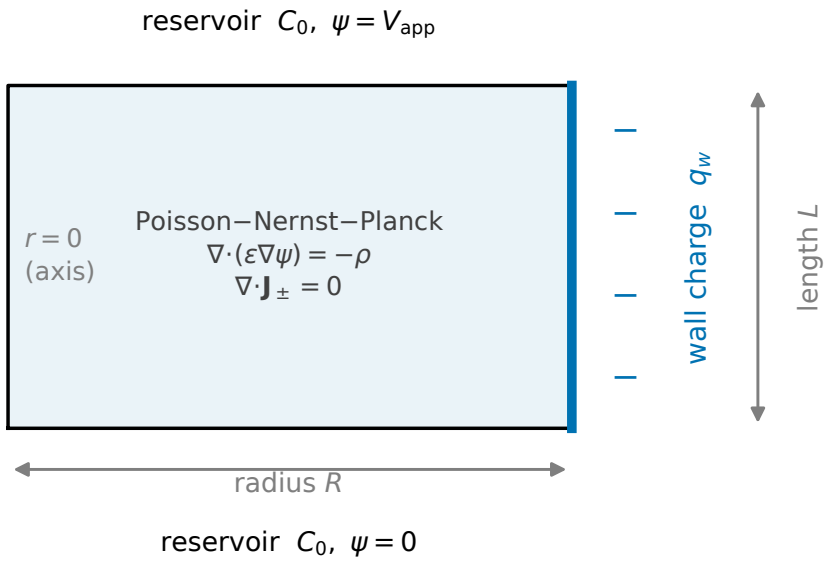
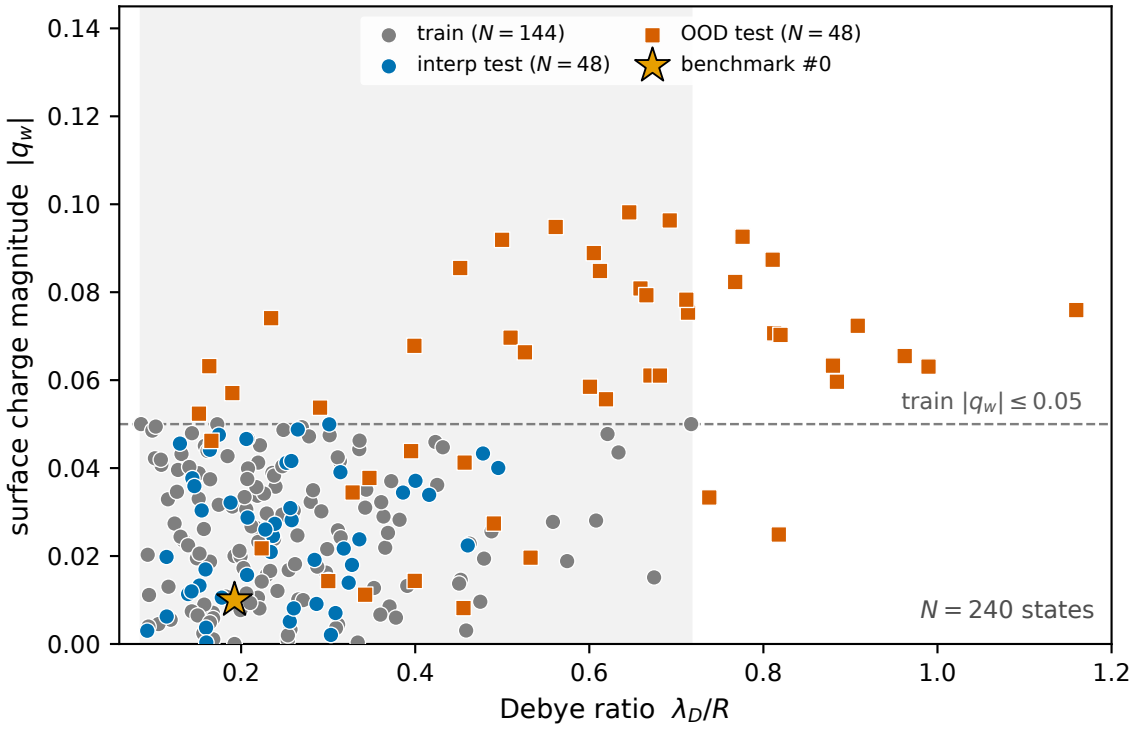
A

axisymmetric charged nanopore



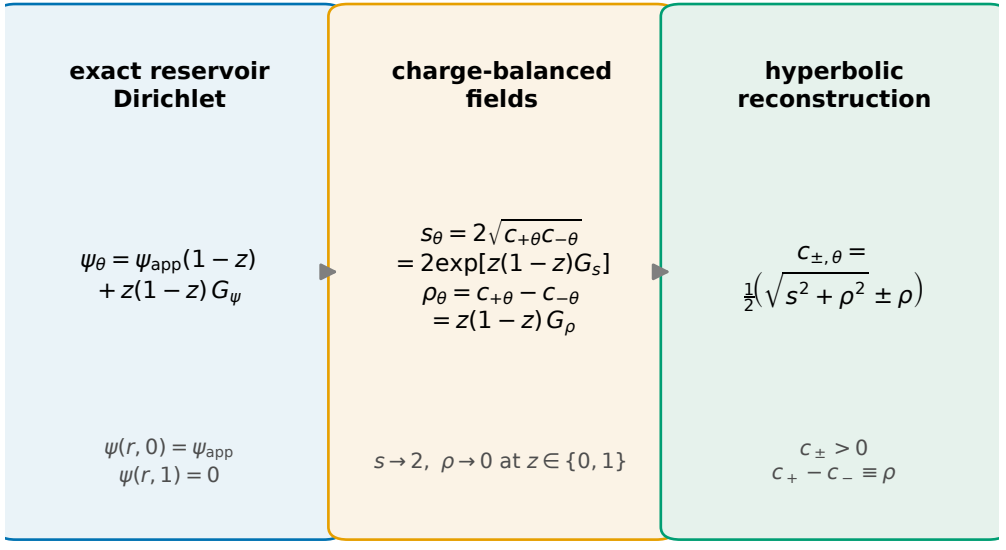
B

4D operating space $\boldsymbol{\mu} = (V_{\text{app}}, C_0, q_w, R)$



C

exact-reservoir charge-balanced ansatz



Weyl singular-value check as sensitivity diagnostic; charge identity is exact

D

parameter-conditioned operator backbones

